

AMBROSE LIGHT, NY, USA

WMO: 994100

Lat: 40.460N

Lon: 73.830W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.6	(c) -8.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 21.3	(k) -2.0	(l) 19.6	(m) 0.5	(n) 11.0	(o) 320	(p) 0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 3.9	(c) 28.9	(d) N/A	(e) 27.2	(f) N/A	(g) 25.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.9	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.6	(b) 17.3	(c) 15.3	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 19.6	(b) 17.3	(c) 15.3	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	0.8	1.1	4.4	9.9	14.2	19.9	22.8	22.6	19.8	14.6	9.1	4.3		
	DBStd	8.63	5.60	4.56	3.85	3.74	3.22	2.76	2.22	2.28	2.25	3.31	3.99	4.00		
	HDD10.0	1008	288	252	177	44	2	0	0	0	0	3	62	179		
	HDD18.3	2710	545	484	431	254	135	15	0	1	12	123	276	435		
	CDD10.0	1744	2	1	3	42	132	297	398	391	294	147	36	3		
	CDD18.3	405	0	0	0	2	7	61	139	133	56	8	0	0		
	CDH23.3	1228	0	0	0	8	29	175	498	449	55	14	0	0		
	CDH26.7	199	0	0	0	1	7	31	88	68	3	1	0	0		
(14) Wind	WSAvg	7.9	9.2	8.8	8.4	8.9	7.9	6.8	6.3	6.1	7.0	7.7	8.6	9.1		
(15) Precipitation	PrecAvg	1164	88	76	106	99	96	100	94	115	97	99	86	109		
	PrecMax	1459	207	148	224	222	202	229	166	341	228	386	147	189		
	PrecMin	926	40	26	27	40	33	22	25	18	35	8	33	34		
	PrecStd	146	34	32	45	44	35	51	38	63	48	74	32	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.1	17.6	24.5	26.3	29.8	31.2	31.6	26.9	25.1	19.1	14.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	11.7	10.4	12.9	20.3	22.8	27.1	29.0	28.4	24.9	22.0	17.1	12.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	10.0	8.4	10.6	17.5	20.3	25.3	27.3	26.7	23.7	20.4	15.8	11.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	8.2	6.6	8.9	14.8	18.6	23.6	25.7	25.6	22.9	19.3	14.9	9.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	5.4	5.2	5.6	5.1	4.7	3.9	3.6	3.6	4.4	4.7	5.0		
		MCDBR	6.9	7.2	7.9	9.9	8.3	7.7	6.4	5.4	4.6	5.4	5.9	6.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.331	0.361	0.385	0.431	0.485	0.490	0.476	0.402	0.365	0.335	0.320		
	taud		2.479	2.456	2.413	2.378	2.281	2.170	2.168	2.199	2.379	2.485	2.524	2.529		
	Ebn at Noon		839	885	890	889	850	802	794	794	836	834	817	813		
	Edh at Noon		77	91	106	117	132	148	147	139	109	88	74	69		
(44) All-Sky Solar Radiation	RadAvg		1.88	2.79	3.79	4.88	5.42	5.96	6.03	5.32	4.38	3.07	2.15	1.63		
	RadStd		0.16	0.29	0.33	0.47	0.56	0.55	0.29	0.34	0.39	0.30	0.18	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+68

Nomenclature: See separate page